

Quantitative modelling student

Aspiring to the research in applied mathematics with a solid foundation in analysis, numerical methods, and mathematical modeling; I am passionate about applying mathematics to address real-world challenges. I thrive in collaborative environments that encourage innovation and continuous learning, aiming to contribute to impactful research and practical solutions in scientific and industrial contexts.

Education

M.Sc., Quantitative Modelling (Ongoing)

UM6P/ IST&I-College of Computing

09/2023 – 09/2025 Morocco

[Semester 3 Capstone Project](#) Integration of Physics-Informed Neural Networks in Alternating Schwarz Methods (December to January 2025).

- Achievements:** Developed PINNs for the modified Helmholtz equation and integrated them with the Alternating Schwarz Method. Optimized convergence using Robin transmission conditions, reducing iterations by 40–50%. Demonstrated scalability on 1D/2D PDEs.

- Tools:** Simlab, Pytorch, VS Code, Kaggle, Python

[Semester 2 Capstone Project](#) Markov Decision Processes (June to July 2024).

- Achievements:** Solved MDPs, implemented Value Iteration, Policy Iteration, Linear Programming. Simulation of the complexity of these methods.

- Tools:** VS Code, Colab, Cplex solver, Python

[Academic Project 1 \(HPC\)](#) Solving 2D Poisson Equation using MPI (March 2024).

- Achievements:** Built a parallel 2D Poisson solver using MPI Cartesian topologies and Finite Difference Method. Implemented domain decomposition and optimized MPI communication.

- Tools:** Python, OpenMPI, NumPy, Matplotlib.

[Academic Project 2 \(HPC\)](#) Parallel Stochastic Gradient (March 2024).

- Achievements:** Built a parallel Stochastic Gradient optimizer using MPI.

- Tools:** Python, OpenMPI, NumPy.

[Academic Project 3](#) Finite Volume Schemes for Advection and Burgers Equations (December 2024).

- Achievements:** Implemented, validated, and compared finite volume schemes (Lax-Friedrichs, Upwind, Lax-Wendroff, Roe) for linear and nonlinear PDEs. Conducted L2 error analysis against exact solutions.

- Tools:** Python, Jupyter Notebook, NumPy, Matplotlib

[Academic Project 4](#) Solving the ODE of Departure Points in Backward Semi-Lagrangian Methods (January 2025)

- Achievements:** Programmed a Runge-Kutta solver to compute departure points, validated results with analytical solutions, performed error analysis, and visualized velocity fields and departure points.

M.Sc. Fundamental Mathematics and Applications (Graduated)

Institute of Mathematics and Physical Science(IMPS)

2021 – 2023 Benin

[Thesis](#): Asymmetric Normed Vector Spaces: First Steps in Asymmetric Functional Analysis.

[Capston Project](#): Lotka–Volterra predator–prey model (November to December 2022)

- Summary:** Implemented schemes, analyzed stability, compared numerical and exact solutions.
- Tools:** Python, VS Code, Jupyter Notebook.

B.Sc. Mathematical Science

Institute of Mathematics and Physical Science(IMPS)

2018 – 2021 Benin

[Thesis](#): Projective real spaces: Some geometrical and topological properties.

Preparatory classes| Mathematics,Physics, and Engineering Sciences

Institute of Mathematics and Physical Science(IMPS)

2018 – 2020 Benin

Work experience

Research Intern (Ongoing)

IST&I-College of Computation, UM6P

02/2025 Morroco, Benguerir

Model Reduction for Partial Differential Equations with Lorentz Cone Constraints:

- Get familiarized with model reduction of PDEs and Lorentz cone type constraints.
- Study contact problems between two elastic bodies.
- Get familiarized with parametrized variational inequalities applied to contact mechanics.
- Implement a proof-of-concept using Python.
- Devise a novel model reduction approach for contact problems.

Work experience

●	Home tutor for several students Home schooling in mathematics 2020 – 2023	Benin
●	Lecturer Mathematics Private school for general education BAMBINO VILLAGE 2021 – 2023	Benin
●	Statistical mathematician intern CaSEG 08/2022 – 09/2022	Lome Togo

Skills & Tools

Frameworks

CVXOPT, OpenFOAM, PyTorch, Pandas, DeepXDE, CUDA, MPI.

Programming

C, Python, JavaScript, R, SQL.

Platforms

Paraview, Medit, Jupyter, VS Code, PyCharm

Tools

FreeFEM++, LATEX, Camtasia, CPLEX .

Leadership & Volunteering

Scout Unit Leader (since 2021)

- Led and organized activities for a group of young scouts (ages 7-11),
- Managed volunteers, coordinated events,
- Strengthened leadership, communication, and conflict resolution skills.

Africitizen

Help recruit talent for the Africiten project

Certification and Competitions

- April 2024, **NVIDIA**: [Fundamentals of Accelerated Computing with CUDA Python](#).
- November 2021, **Udemy**: [Complete web development training](#).
- July 2024, **Math&Maroc Competition for university students** (Morroco).
- June 2015, **National Maths Olympiad for secondary school students** (Benin).
- July 2025, **Smart Farming Hackathon Africa 2025 – 1st Place** (Morroco);
Awarded **1st place** out of 25 teams for innovative, scalable solution at a continent-wide hackathon.

Languages

French

Mother tongue | DALF C1

English

IELTS B2

References

Prof **Imad ELMAHI** | Email : imad.elmahi@um6p.ma

Prof **Amina BENACEUR** | Email : Amina.BENACEUR@um6p.ma

Prof **Khalil FERRADI** | Email : Khalil.FERRADI@um6p.ma

Hobbies

Football, Cardio-Training, Relaxing music, Reading.

Assets

Conscientious, Dilligent, Resilient.